Chapter 10. Nonparametric Inference

Wilcoxon rank-sum and signed-rank tests

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10.1 Introduction

Motivation. With outliers or heavy tails, t methods lose power; sign and rank methods are safer.

Most classical procedures assume the population is normal. If $X_1, \ldots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1),$$

and this pivot gives tests and intervals for μ, approximately valid for non-normal populations by the CLT.

With outliers or far-from-normal populations, normal-theory methods can fail badly: actual confidence below nominal, low power, needlessly long intervals.

Definition (Parametric, nonparametric and distribution-free methods)

(a) A **parametric method** assumes a specific parametric family (e.g. normal) for the population and makes inference about its parameters (μ, σ^2).
(b) A **nonparametric method** assumes no parametric form, typically only that the distribution is *continuous* (or *continuous and symmetric*).
(c) A method valid for every population distribution is **distribution-free**; e.g. a test whose statistic has a null distribution free of the population distribution.

The basic tools are the **sign** (1 if positive, 0 if negative) and the **rank** (position in increasing order) of an observation.

For example, the five observations

$$X_1 = 15, \quad X_2 = 18, \quad X_3 = 11, \quad X_4 = 36, \quad X_5 = 14$$

have ranks

$$R_1 = 3, \quad R_2 = 4, \quad R_3 = 1, \quad R_4 = 5, \quad R_5 = 2.$$

The largest value 36 gets rank 5 however large it is, so working with $R_1, \ldots, R_n$ instead of $X_1, \ldots, X_n$ limits the influence of outliers.

10.2 Wilcoxon rank-sum test

Motivation. The Wilcoxon rank-sum test compares two locations using only the ranks of the pooled sample.

Let the distribution functions be $F(x)$ and $G(x) = F(x - \Delta)$: same shape, shifted by the **shift parameter** $\Delta = \mu_Y - \mu_X$.

Definition (Data structure of the two-sample location problem)

$X_1, X_2, \ldots, X_m$: a random sample from a continuous distribution $F(x)$.
$Y_1, Y_2, \ldots, Y_n$: a random sample from the continuous distribution $F(x - \Delta)$.
The two samples are independent.

$\Delta = 0$ means that X and Y have the same distribution. $\Delta > 0$ means that the center of Y lies to the right of the center of X, and $\Delta < 0$ and $\Delta \neq 0$ are read in the same way.

Definition (Hypotheses on the shift parameter)

(i) $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$.
(ii) $H_0 : \Delta = 0$ versus $H_1 : \Delta < 0$.
(iii) $H_0 : \Delta = 0$ versus $H_1 : \Delta \neq 0$.

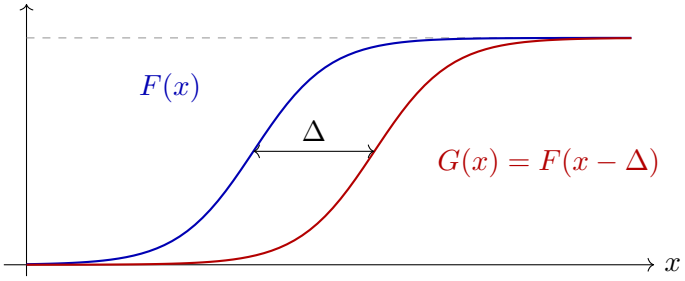


Figure 10.1: The distribution functions F and G under $H_1 : \Delta > 0$.

Pool the X_i and Y_j into the **combined sample** of size $N = m + n$. Under $H_1 : \Delta > 0$ the Y's tend to get larger ranks, so a large Y-rank sum supports H_1.

Definition 10-1 (Wilcoxon rank-sum statistic)

Let R_j be the rank of Y_j in the combined sample of the X's and Y's. The **Wilcoxon rank-sum statistic** is

$$W = \sum_{j=1}^{n} R_j.$$

10.2.1 Null distribution of W

Take $m = 3$, $n = 2$. Under $H_0 : \Delta = 0$, $\{X_1, X_2, X_3, Y_1, Y_2\}$ is a random sample of size 5 from F; with no ties, all $\binom{5}{2} = 10$ orderings of three x's and two y's are equally likely.

No.	1	2	3	4	5	6	7	8	9	10
Arrangement	*yyxxx*	*yxyxx*	*yxxyx*	*yxxxy*	*xyyxx*	*xyxyx*	*xyxxy*	*xxyyx*	*xxyxy*	*xxxyy*
W	3	4	5	6	5	6	7	7	8	9

w	3	4	5	6	7	8	9	Total
$P(W = w)$	0.1	0.1	0.2	0.2	0.2	0.1	0.1	1.0

Tables 10-1 and 10-2: the ten arrangements (each with probability 0.1) and the null distribution of W for $m = 3$, $n = 2$.

The null distribution of W does not involve F, so the rank-sum test is distribution-free. For instance, with $m = 3$, $n = 2$ and $\alpha = 0.10$, the rejection region for $H_1 : \Delta > 0$ is $W \geq 9$, whatever F is.

Tables (Appendix C of the book) or software give the null distribution; the smaller sample is the Y sample, with rank sum W_s, and the table lists $P(W_s \geq x)$ and $P(W_s \leq x^*)$. E.g. for $n = 3$, $m = 7$, $x = 25$, $x^* = 8$:

$$P(W_s \geq 25) = 0.033 = P(W_s \leq 8).$$

(The null distribution is symmetric about $n(N + 1)/2 = 16.5$, so $x^* = n(N + 1) - x$.)

Definition (Wilcoxon rank-sum test)

Take the larger sample as the X sample and the smaller one as the Y sample, and let

$$W = \text{sum of the ranks of the } Y \text{ sample in the combined sample.}$$

The rejection regions for $H_0 : \Delta = 0$ at level α are as follows.
(i) For $H_1 : \Delta > 0$: reject if $W \geq c$.
(ii) For $H_1 : \Delta < 0$: reject if $W \leq c^*$.
(iii) For $H_1 : \Delta \neq 0$: reject if $W \geq c_1$ or $W \leq c_1^*$.
The critical values satisfy

$$P(W \geq c) = \alpha, \qquad P(W \leq c^*) = \alpha, \qquad P(W \geq c_1) = \frac{\alpha}{2} = P(W \leq c_1^*).$$

Example 10-1 (Fine dust in two cities)

Cities A and B have similar population density and geography. Their monthly mean PM_{10} concentrations ($\mu g/m^3$) from March to November were recorded.

Month	3	4	5	6	7	8	9	10	11
City A	33	28	23	25	18	17	25.5	27	26
City B	22	18.5	19.5	20	16	24	22.5	21.5	21

At the 5% level, use the rank-sum test to decide whether the two cities differ.
Solution. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, with $X = A$ and $Y = B$ ($m = n = 9$). The ranks in the combined sample of 18 values are listed below.

X_i	33	28	23	25	18	17	25.5	27	26
rank	18	17	11	13	3	2	14	16	15
Y_j	22	18.5	19.5	20	16	24	22.5	21.5	21
rank	9	4	5	6	1	12	10	8	7

Hence $W_s = 9 + 4 + 5 + 6 + 1 + 12 + 10 + 8 + 7 = 62$. From the table with $m = n = 9$,

$$P(W_s \geq 109) = 0.02 = P(W_s \leq 62).$$

The two-sided p-value is $2 \times 0.02 = 0.04 < 0.05$. We reject H_0 and conclude that the PM_{10} levels of the two cities differ.

Example 10-2 (Tire wear)

Tires of company A cost about 20% more than those of company B. Eight tires of each company were driven 10,000 km and their wear rates (%) were measured. One A measurement was lost.

Company A	15	22	27	18	13	19	25	
Company B	20	16	17	26	28	24	30	33

Can we say that tires of A are better than those of B?
Solution. The sample from A is smaller, so A is the Y sample ($n = 7$) and B is the X sample ($m = 8$). Better quality means lower wear, so we test $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$.

X_i (B)	16	17	20	24	26	28	30	33
rank	3	4	7	9	11	13	14	15
Y_j (A)	13	15	18	19	22	25	27	
rank	1	2	5	6	8	10	12	

So $W_s = 44$ and $(m = 8, n = 7)$ $P(W_s \le 44) = 0.095$: not significant at 5%, but at 10% tires of A are better.

10.2.2 Ties and the normal approximation

Rounded data produce ties; tied observations get the average of their ranks (**midranks**), and the tabulated null distribution then holds only approximately.

When m and n are too large for the tables, we use a normal approximation. We standardize W by its null mean and variance and refer the result to $N(0,1)$, which is a Z test.

Theorem 10-1 (Normal approximation of W)

Under $H_0 : \Delta = 0$,

$$E(W) = \frac{n(m+n+1)}{2}, \qquad \mathrm{Var}(W) = \frac{mn(m+n+1)}{12}.$$

When m and n are large,

$$Z_W = \frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Write $N = m + n$ and assume there are no ties.

Step 1. Under H_0 the N observations are i.i.d. from a continuous distribution. Every ordering of them is therefore equally likely, and the vector $(R_1, \dots, R_n)$ of Y-ranks is a random ordered selection of n values drawn without replacement from $\{1, \dots, N\}$. In particular each R_j is uniform on $\{1, \dots, N\}$, and each pair (R_j, R_k) with $j \neq k$ is uniform on the ordered pairs of distinct values.

Step 2. For a uniform R on $\{1, \dots, N\}$,

$$E(R) = \frac{1}{N}\sum_{r=1}^{N} r = \frac{N+1}{2},$$

$$\mathrm{Var}(R) = \frac{1}{N}\sum_{r=1}^{N} r^2 - \Big(\frac{N+1}{2}\Big)^2 = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4} = \frac{N^2-1}{12}.$$

Step 3. By exchangeability $c = \mathrm{Cov}(R_j, R_k)$ is the same for all $j \neq k$. If all N observations are ranked, the ranks sum to the constant $N(N+1)/2$, whose variance is 0:

$$0 = N\,\frac{N^2-1}{12} + N(N-1)\,c, \qquad \text{so} \qquad c = -\frac{N+1}{12}.$$

Step 4. Hence $E(W) = n(N+1)/2 = n(m+n+1)/2$ and

$$\begin{aligned}\mathrm{Var}(W) &= n\,\frac{N^2-1}{12} - n(n-1)\,\frac{N+1}{12} = \frac{n(N+1)}{12}\,[(N-1) - (n-1)] \\ &= \frac{n(N+1)(N-n)}{12} = \frac{mn(m+n+1)}{12}.\end{aligned}$$

Step 5. W is the total of a simple random sample of size n drawn without replacement from the finite population $\{1,\dots,N\}$. By the central limit theorem for sampling without replacement (Wald–Wolfowitz, Hájek), such a total is asymptotically normal when $\min(n,m)\to\infty$, because no single population value dominates the population variance. Therefore $Z_W \xrightarrow{d} N(0,1)$. ■

Example 10-3 (Normal approximation for Example 10-2)

Use the normal approximation of W for the data of Example 10-2.
Solution. There $W = 44$ with $m = 8$ and $n = 7$, so

$$E(W) = \frac{7(8+7+1)}{2} = 56, \qquad \mathrm{Var}(W) = \frac{8\times 7\times(8+7+1)}{12} = 74.67,$$
$$Z_W = \frac{44-56}{\sqrt{74.67}} = -1.39.$$

$P(Z \le -1.39) = 0.0823$, so the one-sided p-value is 8.23%. This is very close to the exact result of Example 10-2.

Example 10-4 (New surgical method)

A new surgical method A is compared with the standard method B. Each method was applied to 10 patients, and the recovery times (days) are given below.

A	8	9	11	12	16	17	19	19	21	45
B	13	16	18	18	20	20	21	23	25	72

To decide whether A is more effective than B, carry out
(a) the t test;
(b) the Wilcoxon rank-sum test;
(c) the normal-approximation rank-sum test.
Solution. Shorter recovery means a more effective method. With B as the Y sample, we test $H_0 : \Delta = 0$ against $H_1 : \Delta > 0$.
(a) For A, $\bar{x} = 17.7$ and $\sum(x_i - \bar{x})^2 = 1010.1$. For B, $\bar{y} = 24.6$ and $\sum(y_i - \bar{y})^2 = 2600.4$. Then

$$s_p^2 = \frac{1010.1 + 2600.4}{18} = 200.6, \qquad s_p = 14.16,$$
$$t = \frac{\bar{y} - \bar{x}}{s_p\sqrt{1/m + 1/n}} = \frac{24.6 - 17.7}{14.16\sqrt{1/10 + 1/10}} = 1.09.$$

Since $t_{0.1}(18) = 1.330 > 1.09$, the p-value exceeds 10%, and the t test cannot conclude that A is more effective.
(b) With midranks for the ties, the ranks in the combined sample are as follows.

X_i (A)	8	9	11	12	16	17	19	19	21	45
rank	1	2	3	4	6.5	8	11.5	11.5	15.5	19
Y_j (B)	13	16	18	18	20	20	21	23	25	72
rank	5	6.5	9.5	9.5	13.5	13.5	15.5	17	18	20

So $W_s = 128$. With $m = n = 10$ the table gives $P(W_s \ge 128) = 0.045$. At the 5% level, A is more effective than B.
(c) We have

$$E(W) = \frac{10\times 21}{2} = 105, \qquad \mathrm{Var}(W) = \frac{10\times 10\times 21}{12} = 175,$$

$$Z_W = \frac{128 - 105}{\sqrt{175}} = 1.74,$$

and $P(Z \geq 1.74) = 0.0409$. The p-value is 4.09%, so again A is more effective than B.

Why the t test failed here

In Example 10-4 the rank-sum test gives $p \approx 4$–4.5% but the t test $p > 10\%$, because the outliers 45 and 72 inflate s_p. If they were errors or exceptional patients and removed, then

$$\bar{x} = 14.67, \quad \sum(x_i - \bar{x})^2 = 182.00, \qquad \bar{y} = 19.33, \quad \sum(y_i - \bar{y})^2 = 104.00,$$
$$s_p^2 = \frac{182.00 + 104.00}{16} = 17.88, \qquad s_p = 4.23, \qquad t = \frac{19.33 - 14.67}{4.23\sqrt{1/9 + 1/9}} = 4.957.$$

Since $t_{0.0005}(16) = 4.015$, the p-value is below 0.05%, and A is clearly better. With outliers or markedly non-normal data, the rank-sum test is often more efficient than the t test.

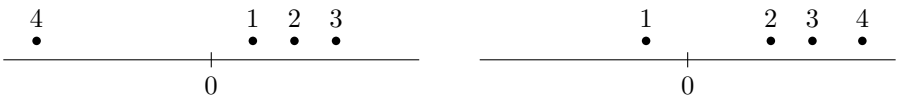
10.3 Wilcoxon signed-rank test

Motivation. The signed-rank test targets the centre of a symmetric distribution, also for paired comparisons.

Definition (Data structure of the one-sample location problem)

$X_1, \ldots, X_n$ is a random sample from $F(x - \theta)$, continuous and symmetric about the **location parameter** θ (mean, if it exists, and median).

Consider two samples of $n = 4$ observations (Figure 10.2). Both have three positive observations. In (a) the negative observation lies far from 0, while in (b) it lies close to 0.



(a) negative value relatively far from 0 (b) negative value relatively close to 0

Figure 10.2: Two samples with $n = 4$; the numbers are the absolute ranks.

The **absolute rank** of an observation is its rank by distance from 0. To test whether the location is 0, use the sum of absolute ranks of the positive observations:

$$\text{(a) } T = 1 + 2 + 3 = 6, \qquad \text{(b) } T = 2 + 3 + 4 = 9.$$

The farther the data are shifted to the right of 0, the larger the ranks or the number of the positive observations, so this sum grows. In general, to test $H_0 : \theta = \theta_0$ we set

$$Z_i = X_i - \theta_0 \qquad (i = 1, \ldots, n).$$

Under H_0, $Z_1, \ldots, Z_n$ follow a distribution symmetric about 0. Let R_i^+ be the absolute rank of $|Z_i|$ among $|Z_1|, \ldots, |Z_n|$. The **Wilcoxon signed-rank statistic** is

$$W^+ = \text{sum of the absolute ranks } R_i^+ \text{ of the positive } Z_i.$$

Under H_0 the absolute ranks of positive and negative Z_i are alike and each sign is $\pm$ with probability $\frac{1}{2}$, so the null distribution of W^+ is free of F: the test is distribution-free.

Appendix C tabulates $P(W^+ \geq x)$ and $P(W^+ \leq x^*)$; for $n = 4$, $P(W^+ \geq 9) = 0.125 = P(W^+ \leq 1)$, since of $2^4 = 16$ sign patterns only $\{2, 3, 4\}$ and $\{1, 2, 3, 4\}$ have rank sum ≥ 9.

Definition (Wilcoxon signed-rank test)

(i) Compute $Z_i = X_i - \theta_0$ for $i = 1, \ldots, n$.
(ii) Find the absolute rank R_i^+ of $|Z_i|$ among $|Z_1|, \ldots, |Z_n|$.
(iii) Compute $W^+ =$ the sum of R_i^+ over the positive Z_i.
The rejection regions for $H_0 : \theta = \theta_0$ are as follows.
For $H_1 : \theta > \theta_0$: reject if $W^+ \geq c$.
For $H_1 : \theta < \theta_0$: reject if $W^+ \leq c^*$.
For $H_1 : \theta \neq \theta_0$: reject if $W^+ \geq c_1$ or $W^+ \leq c_1^*$.
The critical values satisfy

$$P(W^+ \geq c) = \alpha, \quad P(W^+ \leq c^*) = \alpha, \quad P(W^+ \geq c_1) = \frac{\alpha}{2} = P(W^+ \leq c_1^*).$$

Example 10-5 (Smartwatch heart-rate error)

A smartwatch maker claims its heart-rate error (absolute difference from the true rate) averages 5 bpm; a consumer group suspects more. Ten watches on 10 similar people, against a medical monitor, gave these errors (bpm).

3.5	2.6	7.8	6.0	6.8	3.1	8.8	8.5	7.7	4.5

Is the consumer group's claim supported?
Solution. Test $H_0 : \theta = 5$ against $H_1 : \theta > 5$. With $\theta_0 = 5$, the values $Z_i = X_i - 5$ are

$$-1.5,\ -2.4,\ 2.8,\ 1.0,\ 1.8,\ -1.9,\ 3.8,\ 3.5,\ 2.7,\ -0.5.$$

negative Z_i	abs. rank	positive Z_i	abs. rank
-0.5	1	1.0	2
-1.5	3	1.8	4
-1.9	5	2.7	7
-2.4	6	2.8	8
		3.5	9
		3.8	10

So $W^+ = 2+4+7+8+9+10 = 40$. With $n = 10$ the table gives $P(W^+ \geq 40) = 0.116$. The p-value 11.6% is above 5%, so the evidence is insufficient to conclude that the error exceeds 5 bpm.

Ties get average ranks, and the null distribution of W^+ then holds approximately; for large n use the normal approximation.

Theorem 10-2 (Normal approximation of W^+)

Under $H_0 : \theta = \theta_0$,

$$E(W^+) = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}.$$

When n is large,

$$Z_{W^+} = \frac{W^+ - E(W^+)}{\sqrt{\mathrm{Var}(W^+)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Assume there are no ties, and let $Z = X - \theta_0$ have a continuous distribution symmetric about 0 under H_0.

Step 1. The sign of Z is independent of $|Z|$. For $a > 0$, symmetry gives

$$P(Z > 0,\ |Z| \le a) = P(0 < Z \le a) = \tfrac{1}{2}P(|Z| \le a) = P(Z > 0)\,P(|Z| \le a).$$

Step 2. For $k = 1, \dots, n$ let $I_k = 1$ if the observation with absolute rank k is positive, and $I_k = 0$ otherwise. The absolute ranks depend only on $|Z_1|, \dots, |Z_n|$, and by Step 1 the signs are i.i.d. fair coin flips independent of these. Hence $I_1, \dots, I_n$ are i.i.d. Bernoulli$(\tfrac{1}{2})$, and

$$W^+ = \sum_{k=1}^{n} k\, I_k.$$

Step 3. With $E(I_k) = \tfrac{1}{2}$ and $\mathrm{Var}(I_k) = \tfrac{1}{4}$,

$$E(W^+) = \frac{1}{2}\sum_{k=1}^{n} k = \frac{n(n+1)}{4},$$
$$\mathrm{Var}(W^+) = \frac{1}{4}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}.$$

Step 4. The summands $Y_k = k(I_k - \tfrac{1}{2})$ are independent with mean 0 and $|Y_k| = k/2$. Write $s_n^2 = \mathrm{Var}(W^+)$. Then

$$\frac{\sum_{k=1}^{n} E|Y_k|^3}{s_n^3} = \frac{\frac{1}{8}\sum k^3}{s_n^3} = \frac{n^2(n+1)^2/32}{\big(n(n+1)(2n+1)/24\big)^{3/2}} = O(n^{-1/2}) \to 0.$$

This is Lyapunov's condition, so the Lyapunov CLT gives $Z_{W^+} \xrightarrow{d} N(0,1)$. ∎

10.3.1 Paired comparison

For heterogeneous units, homogeneous pairs beat random splitting (Section 6.2). With n pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ form the differences

$$D_i = Y_i - X_i \qquad (i = 1, \dots, n).$$

$H_0 : \Delta = 0$ vs. $H_1 : \Delta > 0$ becomes $H_0 : \theta = 0$ vs. $H_1 : \theta > 0$ for the D_i; the signed-rank test on the D_i is a nonparametric paired comparison.

Example 10-6 (Twins and a new curriculum)

For a new curriculum A, 16 pairs of identical twins with equal scores were recruited; one twin of each pair took the old curriculum B, the other A, then both sat the same test.

A	70	85	73	75	65	80	71	80
B	65	43	45	80	84	71	52	42
$Z = A - B$	5	42	28	−5	−19	9	19	38
A	51	72	76	79	59	84	56	57
B	78	62	38	80	54	87	38	43
$Z = A - B$	−27	10	38	−1	5	−3	18	14

At the 5% level, is the new curriculum more effective?

Solution. Let θ be the location of Z, and test $H_0 : \theta = 0$ against $H_1 : \theta > 0$. Absolute ranks (with midranks for ties) are as follows.

negative Z_i	-1	-3	-5	-19	-27
abs. rank	1	2	4	10.5	12

positive Z_i	5	5	9	10	14	18	19	28	38	38	42
abs. rank	4	4	6	7	8	9	10.5	13	14.5	14.5	16

So $W^+ = 106.5$. With $n = 16$, Theorem 10-2 gives

$$E(W^+) = \frac{16 \times 17}{4} = 68, \qquad \mathrm{Var}(W^+) = \frac{16 \times 17 \times 33}{24} = 374,$$
$$Z_{W^+} = \frac{106.5 - 68}{\sqrt{374}} = 1.99.$$

$P(Z \geq 1.99) = 0.0233$, so the one-sided p-value 2.33% is below 5%. We reject H_0 and conclude that the new curriculum A is more effective than B.